

Continuous probability distribution:

A function with values $f(x)$ defined over the set of all real numbers, is called a probability density function(pdf) of the continuous random variable x if

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$

$$P(-\infty \leq x \leq \infty) = \int_{-\infty}^{\infty} f(x) dx$$

- Properties:

A function can serve as a probability density function of a continuous random variable x if its values of $f(x)$ satisfy the conditions

1. $f(x) \geq 0, -\infty < x < \infty$

2. $\int_{-\infty}^{\infty} f(x) dx = 1$

Cumulative distributive function

If X is a continuous random variable and $F(x)$ is its probability density, then,

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

- Let $f(x)$ be the probability density function of a continuous random variate, then

1. Mean (μ) = $E(X) = \int_{-\infty}^{\infty} x f(x) dx$

2. Median is the point which divides the entire distribution into two equal parts.

3. Mode is the value of x for which $f(x)$ is maximum.

4. Variance(x) = $\sigma^2 = \int_a^b x^2 f(x) dx - \mu^2$

Example:

A continuous random variable x is defined by

$$f(x) = \begin{cases} \frac{1}{16} (3+x)^2, & \text{if } -3 \leq x < -1 \\ \frac{1}{16} (6-2x)^2, & \text{if } -1 < x < 1 \\ \frac{1}{16} (3-x)^2 & \text{if } 1 \leq x < 3 \\ 0, & \text{else where} \end{cases}$$

Verify that $f(x)$ is a density function and also find the mean of x .

Solution:

Here $f(x) \geq 0 \forall x \in [-3,3]$

And

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{-3} f(x) dx + \int_{-3}^{-1} f(x) dx + \int_{-1}^1 f(x) dx + \int_1^3 f(x) dx + \int_3^{\infty} f(x) dx$$

$= 1$

Since $f(x)$ satisfies the requirements for a density function

$\therefore f(x)$ is a density function.

$$\begin{aligned} \text{Mean } (\mu) &= E(X) = \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_{-\infty}^{-3} x f(x) dx + \int_{-3}^{-1} x f(x) dx + \int_{-1}^1 x f(x) dx + \int_1^3 x f(x) dx + \int_3^{\infty} x f(x) dx \end{aligned}$$

$=$

$= 0$

Example:

The trouble shooting capability of an IC chip in a circuit is a random variable X whose distribution function is given by

$$F(X) = \begin{cases} 0, & \text{for } x < 0 \\ 1 - \frac{9}{x^2}, & \text{for } x \geq 0 \end{cases}$$

Where x denote the number of years.

Find the probability that the IC chip will work properly

- (i) Less than 8 years
- (ii) Beyond 8 years
- (iii) Between 5 to 7 years
- (iv) Anywhere from 2 to 5 years.

Solution:

We know $F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$

Here

$$F(X) = \int_0^x f(t)dt = \begin{cases} 0, & \text{for } x < 0 \\ 1 - \frac{9}{x^2}, & \text{for } x \geq 3 \end{cases}$$

$$(i) P(X \leq 8) = \int_0^8 f(t)dt = \left(1 - \frac{9}{8^2}\right) = 0.8594$$

$$(ii) P(X > 8) = 1 - P(X \leq 8) = 1 - 0.8594 = 0.1406$$

$$\begin{aligned} (iii) P(5 \leq x \leq 7) &= F(7) - F(5) \\ &= \left(1 - \frac{9}{7^2}\right) - \left(1 - \frac{9}{5^2}\right) = 0.1763 \end{aligned}$$

$$\begin{aligned} \text{(iv) } P(2 \leq x \leq 5) &= F(5) - F(2) \\ &= \left(1 - \frac{9}{5^2}\right) - 0 = 0.64 \end{aligned}$$

Example:

The daily consumption of electric power(in millions of KW-hours) is a random variable having the probability density function(p.d.f.)

$$f(x) = \begin{cases} \frac{1}{9}x e^{-x/3}, & \text{for } x > 0 \\ 0, & \text{for } x \leq 0 \end{cases}$$

If the total production is 12 million KW-hours, determine the probability that there is power cut (shortage) on any given day.

Solution:

Power supply is inadequate if daily consumption **exceeds 12 million** KW,
i.e., $P(X > 12)$

Probability that the power consumed is between 0 and 12 is

$$\begin{aligned} p(0 \leq x \leq 12) &= \int_0^{12} f(x) dx = \frac{1}{9} \int_0^{12} x e^{-x/3} dx \\ &= \frac{1}{9} \left[x \left(\frac{e^{-x/3}}{(-1/3)} \right) - 1 \cdot \left(\frac{e^{-x/3}}{(1/9)} \right) \right] \quad U.L = 12, L.L. = 0 \\ &= 1 - 5e^{-4} \end{aligned}$$

Power supply is inadequate if daily consumption exceeds 12 million KW,

$$\begin{aligned} P(X > 12) &= 1 - P(0 \leq X \leq 12) \\ &= 1 - (1 - 5e^{-4}) = 0.0915781 \end{aligned}$$

Example:

Find the constant K such that

$$f(x) = \begin{cases} k x^2, & \text{if } 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

Is a probability function.

(i) Find the cumulative probability distribution function $F(x)$.

(ii) $P(1 < X \leq 2)$.

Solution:

$$\text{given } f(x) = \begin{cases} k x^2, & \text{if } 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

since

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= 1 \\ \Rightarrow \int_0^3 k x^2 dx &= 1 \\ k &= \frac{1}{9} \end{aligned}$$

- The distribution function is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$
$$F(x) = \int_0^x \frac{1}{9} t^2 dt = \frac{1}{27} x^3$$

$$\therefore F(x) = \begin{cases} \frac{1}{27} x^3, & \text{if } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

- $P(1 < x \leq 2) = \int_1^2 \left(\frac{1}{9} x^2 \right) dx = \frac{7}{27}$

Practice Problems:

1. The diameter of an electric cable is assumed to be continuous random variable with probability density function

$$f(x) = 6x(1 - x), 0 \leq x \leq 1. \text{ Justify.}$$

Find the mean and variance of the distribution.

$$(\text{answer: Mean}=\frac{1}{2}, \text{ Variance}=\frac{1}{20})$$

Practice problem:

2. The length of time (in minutes) that a certain lady speaks on the telephone is found to be random phenomenon, with a probability function specified by the function

$$f(x) = \begin{cases} A e^{-x/5}, & \text{for } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find the value of A that makes f(x) a probability density function.
- (ii) What is the probability that the number of minutes that she will take over the phone is more than 10 minutes?

$$(\text{answers: } A = \frac{1}{5}, P(X > 10) = \frac{1}{e^2})$$

Practice problem:

3. The probability density $f(x)$ of a continuous random variable is given by $f(x) = c e^{-|x|}$, $-\infty < x < \infty$. Find c and find the mean and variance of the distribution. Also find the probability that the variate lies between 0 and 4.

(answers: $c=1/2$, mean=0, variance=2 and 0.4908)